

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0863

North Tifton 100 kV
March 20, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0863. The Interconnection Customer proposes a 128.1 MW solar photovoltaic facility interconnecting at the North Tifton 100 kV substation in the Indiana area. The request was received on September 14, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 320 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$11,823,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Richard Lake - Savona 100kV	East Methuen 500-230 kV TX	153.0	110	8.6
Savona Substation - Butler Municipal Power Plant 100kV	East Ethuen 500/230kV Auto	151.8	121	8.6
Hansel - Butler Municipal Power Plant 100kV	East Methuen 500/230 kV TX	127.9	155	8.6
Thousand Springs - Blount 100kV line	Powerline Elkhorn - Websterville 100kV	126.7	130	6.1
Richard Lake - Carbondale Nw 100kV	East Methuen 500/230 kV Transformer Bank	125.4	116	8.6
Thousand Springs - Hansel 100kV L	East Methuen 500-230 kV TX	125.4	165	8.6
Snyder - Rancho Santa Fe 100kV line	Powerline Elkhorn Substation - Websterville 100kV	123.6	128	6.1
Rancho Santa Fe - Blount 100kV	Websterville - Powerline Elkhorn 100kV L	120.8	124	6.1
Websterville - Powerline Elkhorn 100kV	E Mehuen 500-230 kV Auto	115.8	134	6.8
Thousand Springs Substation - Powerline Elkhorn Substation 100kV	E Methuen 500 kV/230 kV Transformer	108.5	127	6.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Richard Lake - Savona 100kV	Rebuild line with higher-capacity conductor	\$6,321,000	16% (\$1,029,000)
Savona Substation - Butler Municipal Power Plant 100kV	Rebuild line with higher-capacity conductor	\$2,720,000	16% (\$443,000)
Hansel - Butler Municipal Power Plant 100kV	Rebuild line with higher-capacity conductor	\$5,942,000	16% (\$968,000)
Thousand Springs - Blount 100kV	Rebuild line with higher-capacity conductor	\$2,720,000	28% (\$768,000)
Richard Lake - Carbondale Nw 100kV	Rebuild line with higher-capacity conductor	\$2,255,000	16% (\$367,000)
Thousand Springs - Hansel 100kV L	Rebuild line with higher-capacity conductor	\$9,519,000	16% (\$1,550,000)
Snyder - Rancho Santa Fe 100kV line	Rebuild line with higher-capacity conductor	\$9,241,000	28% (\$2,609,000)
Rancho Santa Fe - Blount 100kV	Rebuild line with higher-capacity conductor	\$3,092,000	28% (\$873,000)
Websterville - Powerline Elkhorn 100kV	Rebuild line with higher-capacity conductor	\$10,214,000	28% (\$2,898,000)
Thousand Springs Substation - Powerline Elkhorn Substation 100kV	Reconductor line	\$1,120,000	28% (\$318,000)
	Total allocated to project		\$11,823,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.